

Introduction to Pandas

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Pandas is the central element of the **Python** ecosystem for *data science*. The recent success of **Python** in data analysis owes a lot to **Pandas**, which made it possible to import **SQL** logic into the **Python** language. **Pandas** embeds a huge number of features that allow efficient processing pipelines for medium-sized data (up to a few gigabytes). Beyond this volume, other solutions will be needed (**DuckDB**, **Dask**, **Polars**, **Spark**...).

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The **Pandas** part has recently evolved. You can find the content related to **Pandas** in the following chapters:

- Introduction to **Pandas** and the logic of the *DataFrame*;
- Combining data with **Pandas**: joins;
- Handling data with **Pandas**: statistics and reshaping;
- Pandas: limits and alternative ecosystems;

Informations additionnelles

i Python environment

This site was built automatically through a **Github** action using the **Quarto** reproducible publishing software (version 1.10.18).

The environment used to obtain the results is reproducible via **uv**. The **pyproject.toml** file used to build this environment is available on the **linogaliana/python-datascientist** repository

```
[project]
name = "python-datascientist"
version = "0.1.0"
description = "Source code for Lino Galiana's Python for data science course"
readme = "README.md"
requires-python = "≥3.13,<3.14"
dependencies = [
    "altair≥6.0.0",
    "cartiflette",
    "contextily=1.6.2",
    "duckdb≥0.10.1",
```

```

    "folium≥0.19.6",
    "gdal=3.11.4",
    "graphviz=0.20.3",
    "great-tables≥0.12.0",
    "gt-extras≥0.0.8",
    "ipykernel≥6.29.5",
    "jupyter≥1.1.1",
    "jupyter-cache≥1.0.0",
    "kaleido≥0.2.1",
    "langchain-community≥0.3.27",
    "loguru=0.7.3",
    "markdown≥3.8",
    "nbcclient≥0.10.0",
    "nbformat≥5.10.4",
    "nltk≥3.9.1",
    "pandas≥3.0",
    "pip≥25.1.1",
    "plotly≥6.1.2",
    "plotnine≥0.15",
    "polars≥1.8.2",
    "pyarrow≥17.0.0",
    "pynsee≥0.1.8",
    "python-dotenv≥1.0.1",
    "python-frontmatter≥1.1.0",
    "pywaffle≥1.1.1",
    "requests≥2.32.3",
    "scikit-image≥0.24.0",
    "scikit-learn≥1.8.0",
    "scipy≥1.13.0",
    "seaborn≥0.13.2",
    "selenium<4.39.0",
    "spacy≥3.8.4",
    "webdriver-manager≥4.0.2",
    "wordcloud=1.9.3",
]

[tool.uv.sources]
cartiflette = { git = "https://github.com/inseefrlab/cartiflette" }
gdal = [
    { index = "gdal-wheels", marker = "sys_platform = 'linux'", },
    { index = "geospatial-wheels", marker = "sys_platform = 'win32'", },
]

[[tool.uv.index]]
name = "geospatial-wheels"
url = "https://nathanjmcDougall.github.io/geospatial-wheels-index/"
explicit = true

[[tool.uv.index]]
name = "gdal-wheels"
url = "https://gitlab.com/api/v4/projects/61637378/packages/pypi/simple"
explicit = true

[dependency-groups]
dev = [
    "nb-clean≥4.0.1",
]

```



To use exactly the same environment (version of `Python` and *packages*), please refer to the documentation for `uv`.

Bibliography